Microsoft AI Tour

Microsoft

Advanced retrieval for your AI Apps and Agents on Azure

Subtitle or speaker name

Most agents need the ability to find relevant context

Zava Shopper Agent

Finds products

Zava Interior Design Agent

Finds design images

Zava Inventory Agent

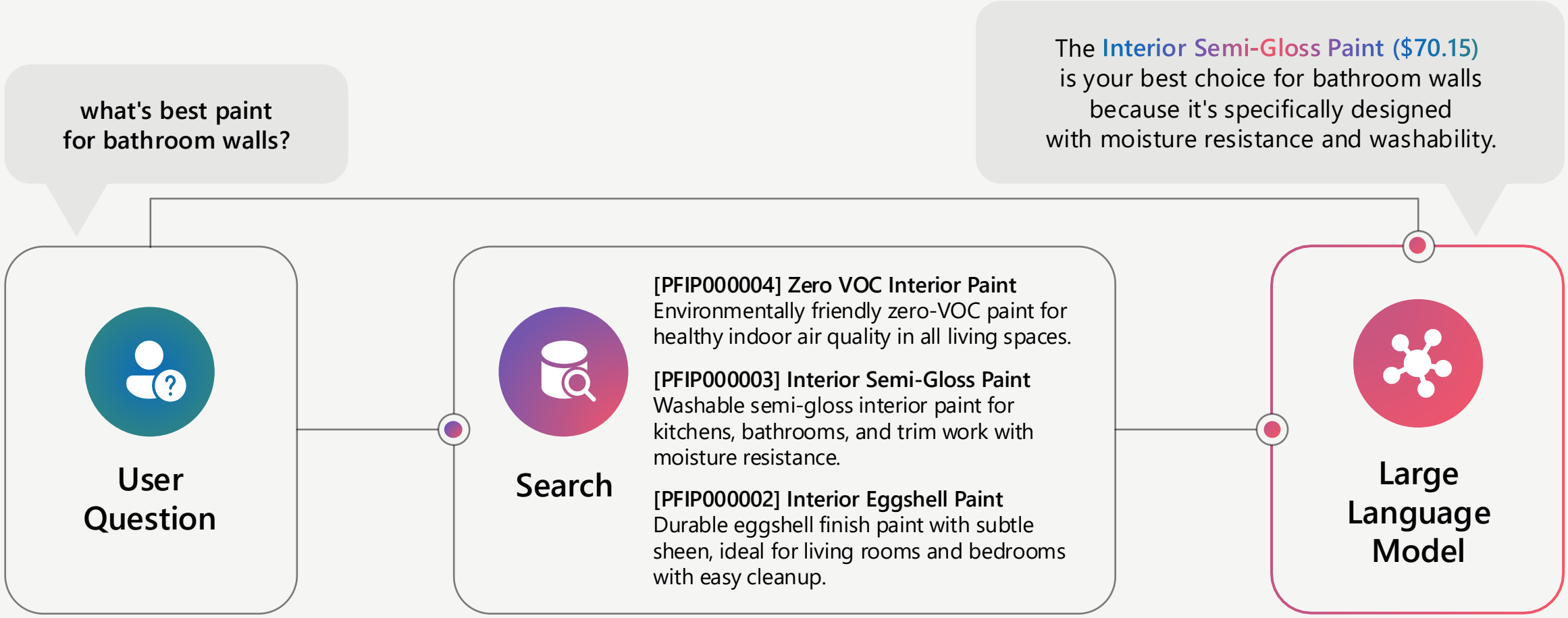
Finds inventory

Zava HR Agent

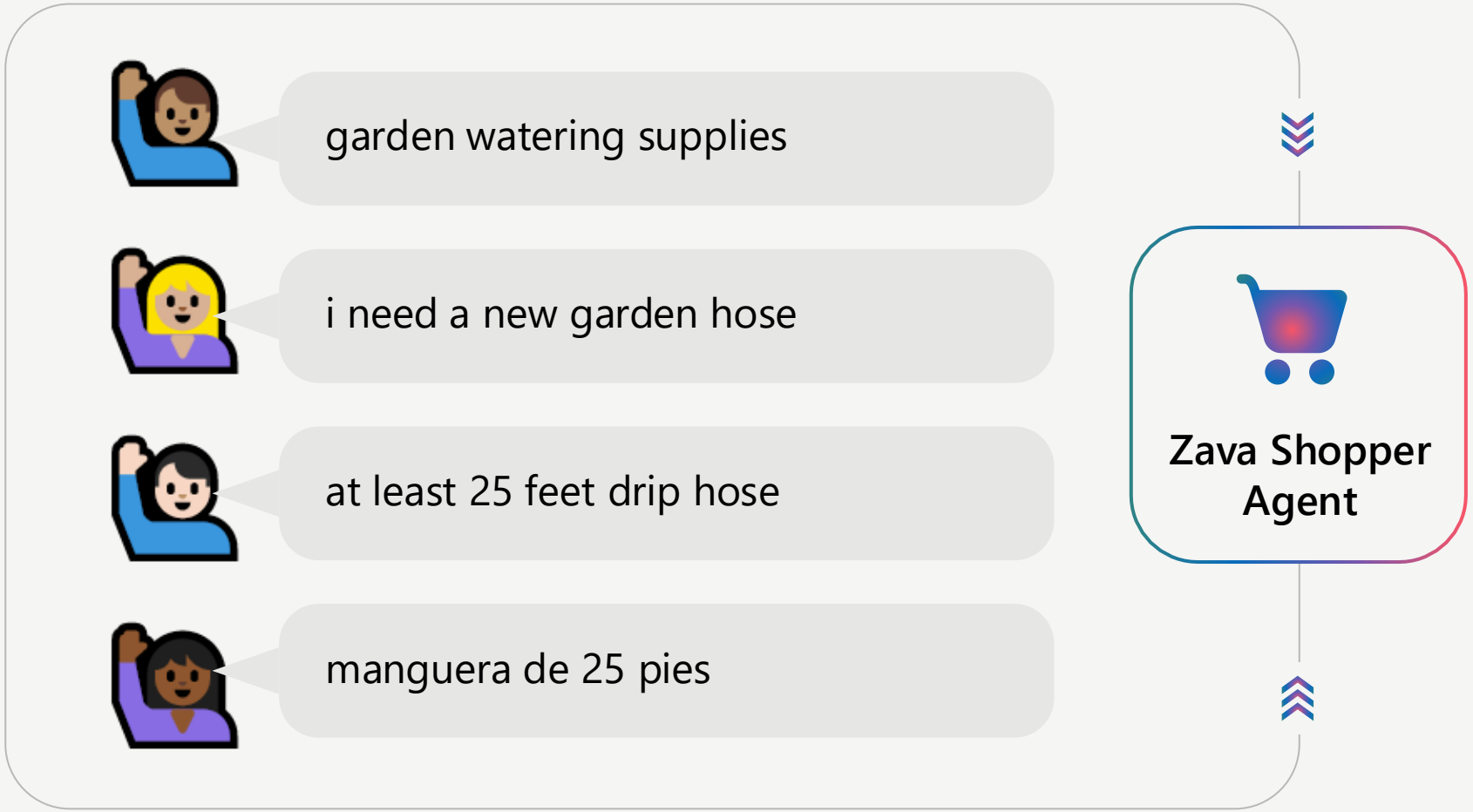
Finds documents

How do agents answer questions? RAG!

Retrieval-Augmented Generation (RAG) grounds the response from an LLM in the retrieved context.



Users need agents to find all sorts of things



Agenda

How do we supercharge retrieval?

Optimal search strategies

Graph-based retrieval with PostgreSQL

Agentic retrieval with Azure AI Search

Optimal search strategies

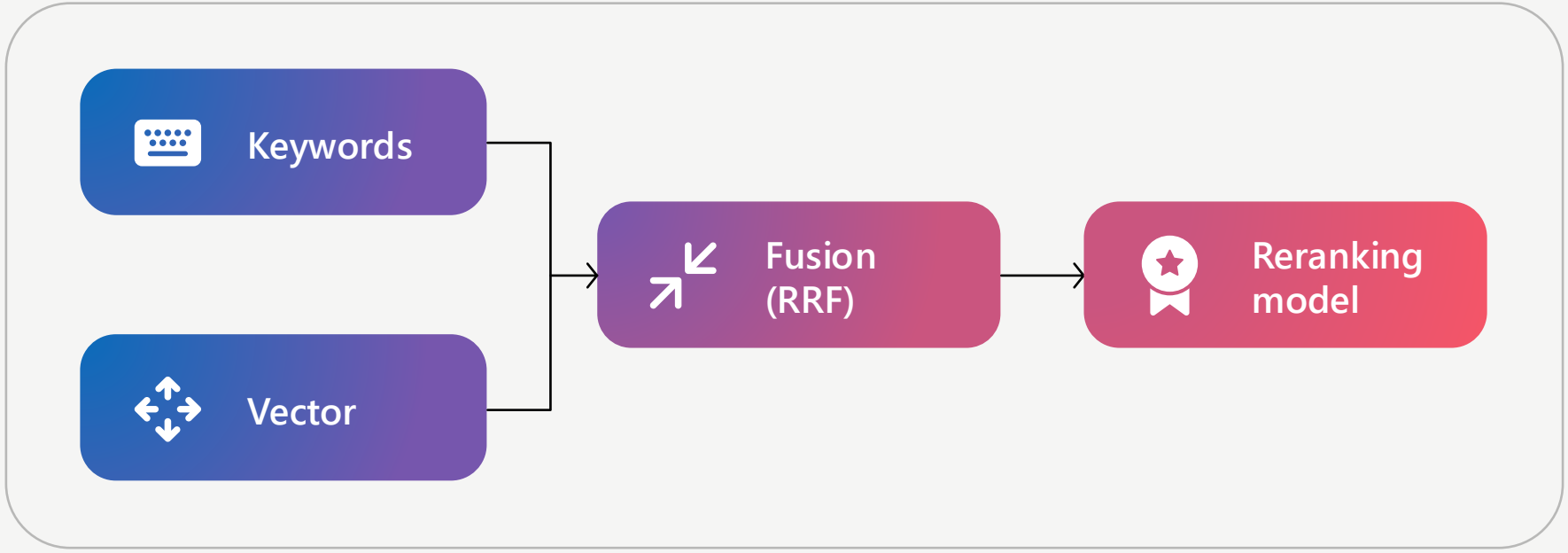
Different queries need different search strategies

Query	Vector search	Keyword search
25 foot hose	✗	✓
25 foot drip hose	✓	✓
garden watering supplies	✓	✗
manguera de jardin	✓	✗

Hybrid search: The best of both worlds!

Complete search stacks do better:

- Hybrid retrieval (keywords + vectors) > pure-vector or keyword
- Hybrid + Reranking > Hybrid



Let's break those down...

Keyword search

Best for: exact phrase matches

Documents are stored in an **inverted index** which can be queried efficiently.

ID	Name	Description
335	Soaker Hose 25-foot	Porous soaker hose for efficient water delivery directly to plant roots.
334	Garden Hose 50-foot	Heavy-duty garden hose with brass fittings and kink-resistant construction.

→

Term	Document List
hose	335, 336
foot	335, 336
garden	335
25	336

"25 foot hose" → **Full text search of name + description** →

ID	Document List	Relevance
335	Soaker Hose 25-foot	0.066
336	Expandable Hose 100-foot	0.046
334	Garden Hose 50-foot	0.046
312	Drain Snake 25-foot	0.040

Demo: keyword_search.py

Tip: **BM25** is the best algorithm, but other inverted index algorithms are also effective.

Vector search

Best for: conceptually similar matches

1. **Encode queries as embeddings** (lists of floating point numbers)
2. **Find semantically similar documents by comparing embeddings**

Query

"garden watering supplies"

Compute embedding vector

Query vector

[-0.003335318, -0.0176891904,...]

Search existing vectors

K closest vectors

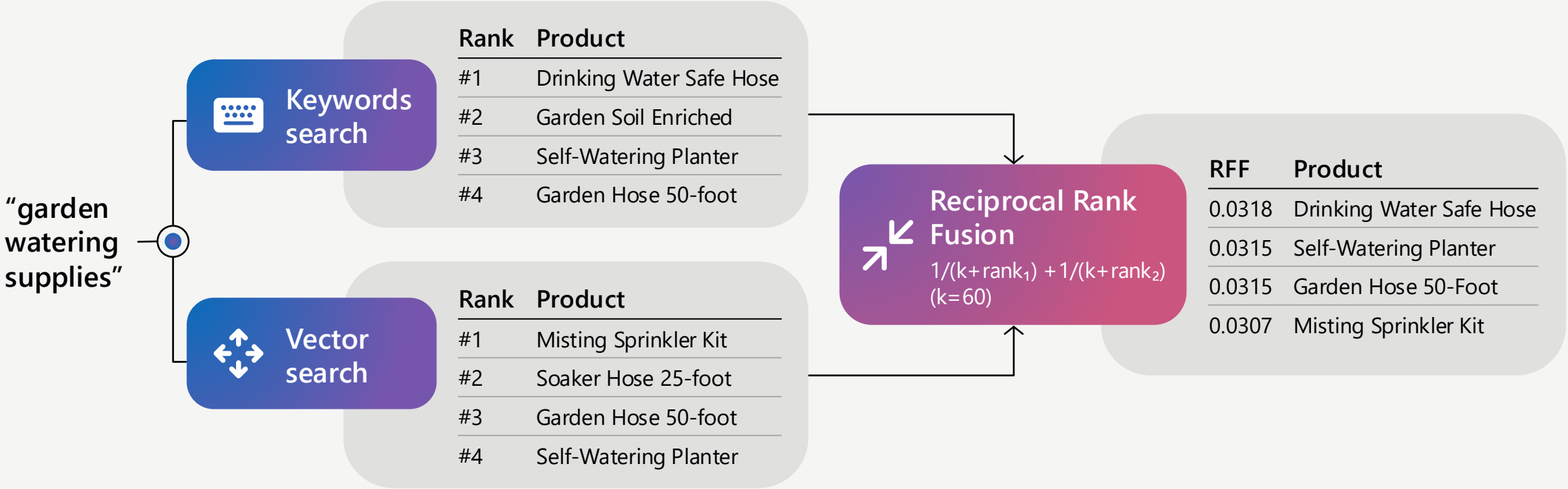
Tip: Use approximate algorithm (**HNSW** or **DiskANN**) for more efficient search.

Demo: vector_search.py

ID	Name	Cosine Similarity
335	Soaker Hose 25-foot	0.51
336	Garden Hose 50-foot	0.51
338	Retractable Hose Hanger	0.46

RRF: Reciprocal Rank Fusion

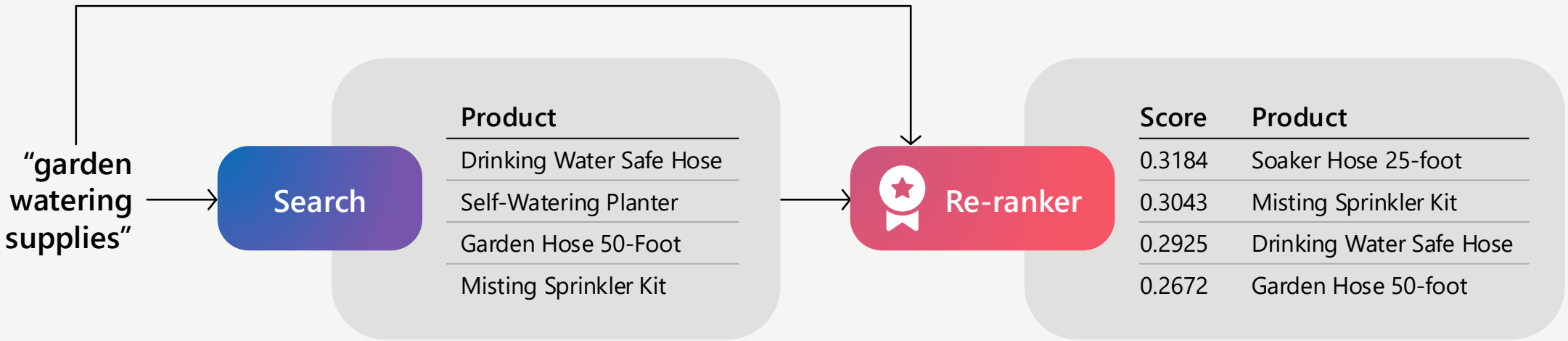
Using text **and** vector search? Merge results using RRF to rank results based off their relative ranks.



Demo: hybrid_rrf_search.py

Re-ranking

A re-ranker model gives documents a relevance score according to the query.

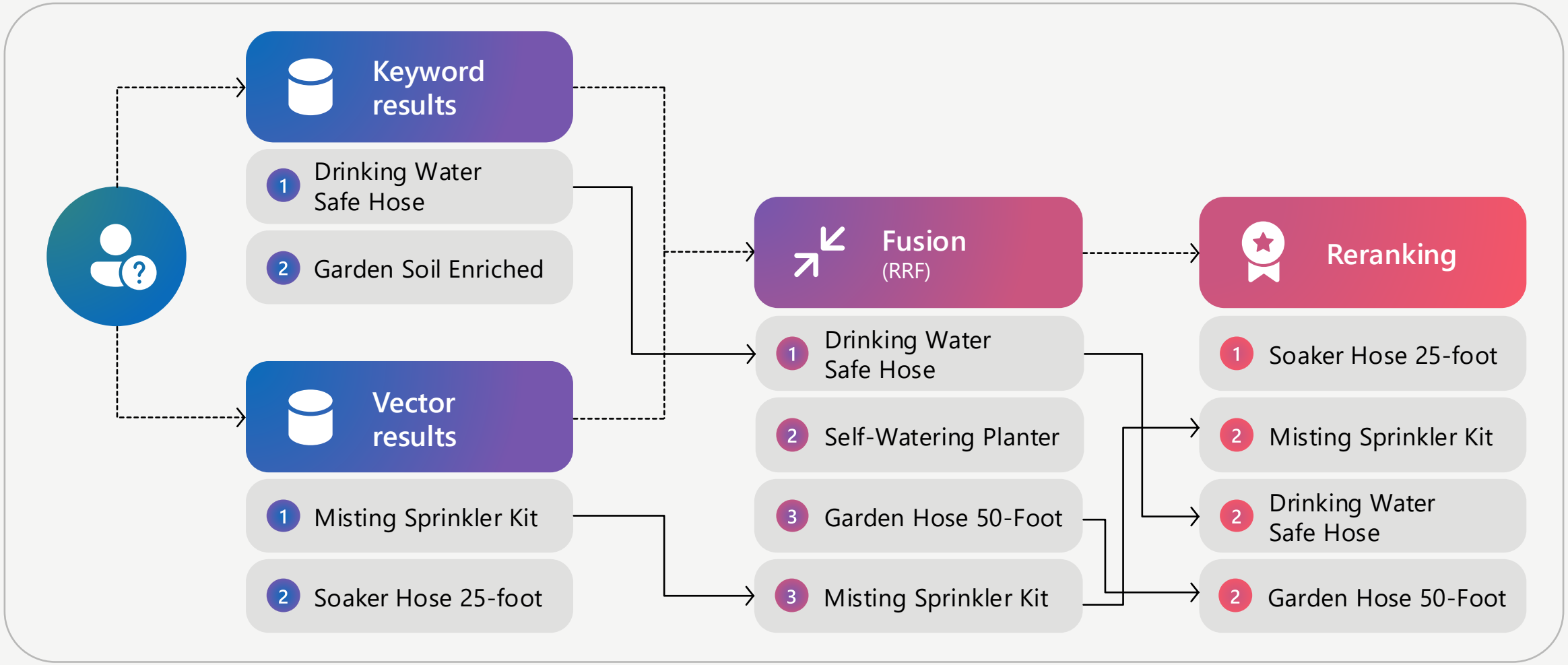


Tip: Cross-encoder models (like Cohere Reranker) are best, but LLMs can be used as well.

Demo: hybrid_ranker_search.py

A complete hybrid search flow

Question: "garden watering supplies"



Hybrid search is supported on many Azure databases

Database	Vector search?	Keyword search?	RRF?	Re-ranker?
Azure Database for PostgreSQL	✓ pgvector HNSW/IVF/DiskANN	✓ fts	✓	✓ azure_ai.rank()
Azure AI Search	✓ HNSW	✓ BM25	✓	✓
Azure Cosmos DB	✓ VectorDistance() DiskANN	✓ BM25	✓	(Private preview)
Azure SQL	✓ vector_distance() DiskANN	✓ BM25	✓	✗

Impact of hybrid search across query types

Query type	Keyword [NDCG@3]	Vector [NDCG@3]	Hybrid [NDCG@3]	Hybrid + Semantic ranker [NDCG@3]
Concept seeking queries	39	45.8	46.3	59.6
Fact seeking queries	37.8	49	49.1	63.4
Exact snippet search	51.1	41.5	51	60.8
Web search-like queries	41.8	46.3	50	58.9
Keyword queries	79.2	11.7	61	66.9
Low query/doc term overlap	23	36.1	35.9	49.1
Queries with misspellings	28.8	39.1	40.6	54.6
Long queries	42.7	41.6	48.1	59.4
Medium queries	38.1	44.7	46.7	59.9
Short queries	53.1	38.8	53	63.9

Source: Outperforming vector search with hybrid + reranking

But... hybrid search isn't always enough!

Queries requiring external knowledge

Do you carry any of the pesticides approved for use in berkeley, CA?

Queries about relations

I want a fairly cheap garden hose that's reviewed as highly as the expensive hoses

Multiple questions in one query

I'm setting up a planter for veggies in part shade. What soil amendments do I need and which veggies grow with only afternoon sunlight?

Agentic Graph RAG with Azure Database for PostgreSQL

Users ask hard questions about relational data

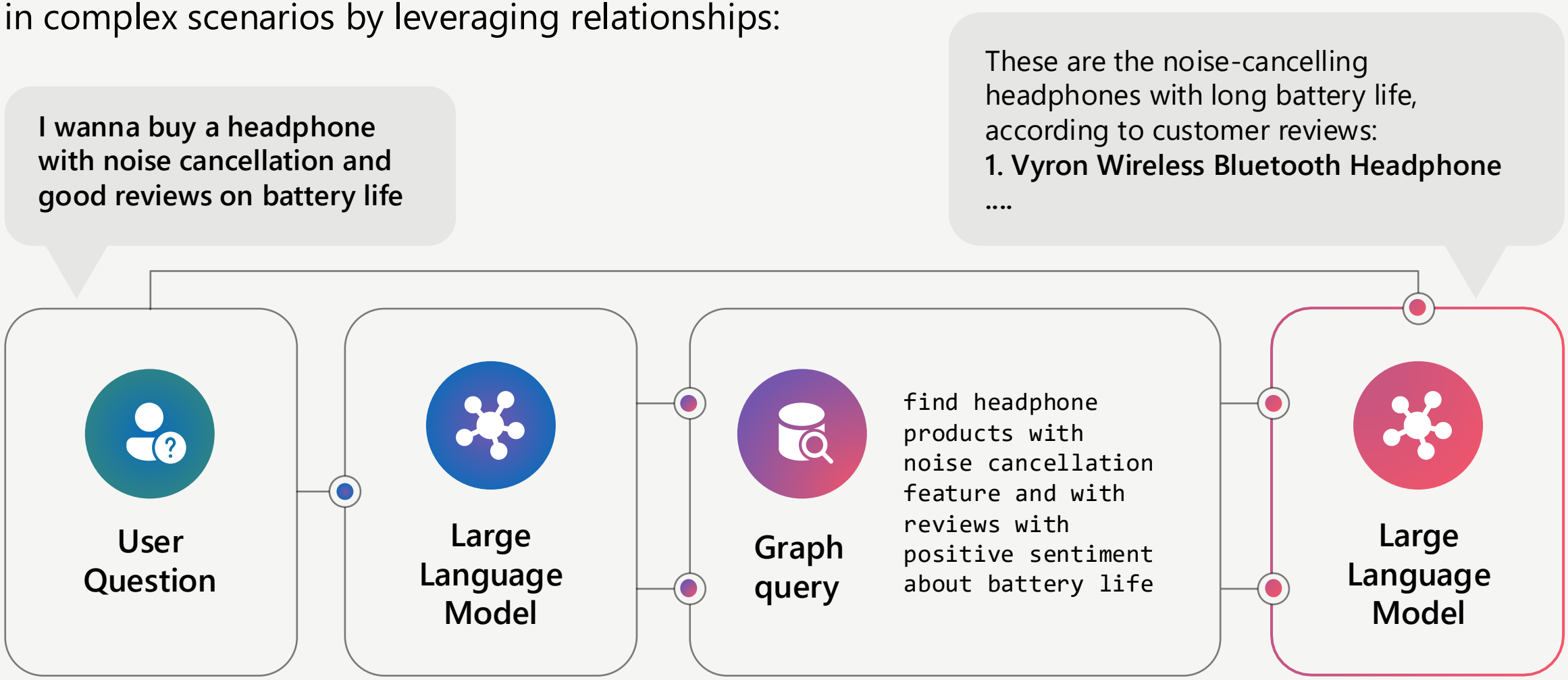
i want a cheap garden hose that's reviewed as highly as the expensive hoses

i want a headphone with noise cancellation and good reviews on battery life

which category of hedge trimmers are reviewed more highly, the electric trimmers or the gas trimmers?

RAG with graph queries

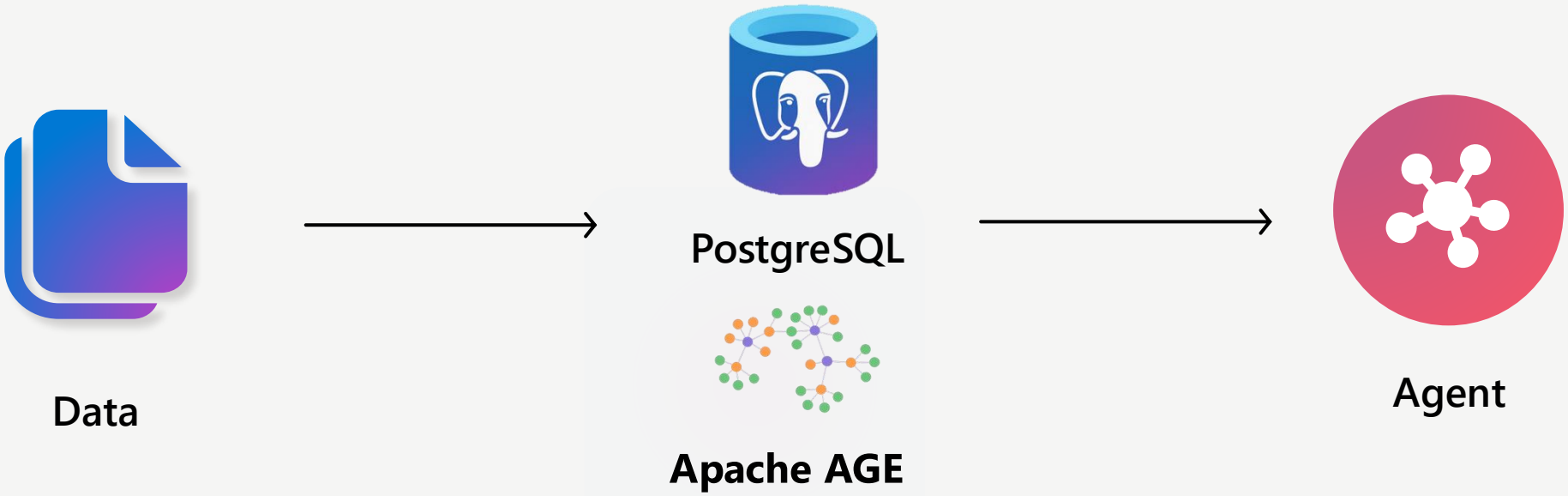
Use graph queries to improve results
in complex scenarios by leveraging relationships:



Apache AGE on Azure Database for PostgreSQL

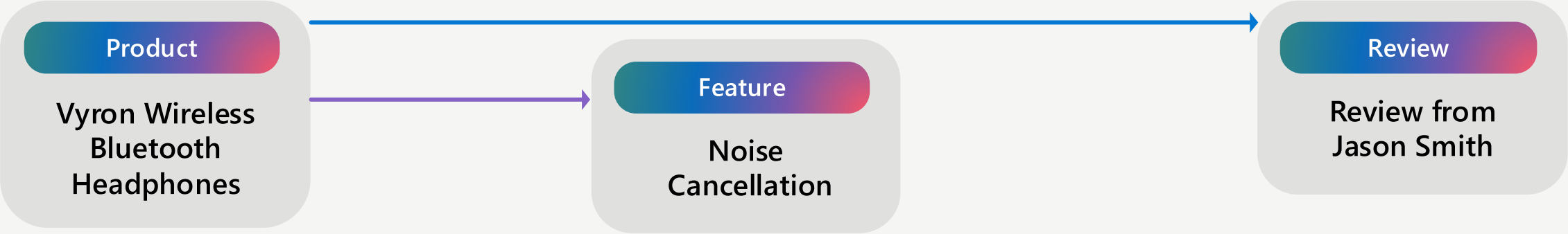
Graph database plugin for PG

Cypher + SQL Hybrid Queries	Fast graph processing	Graph visualization & analytics



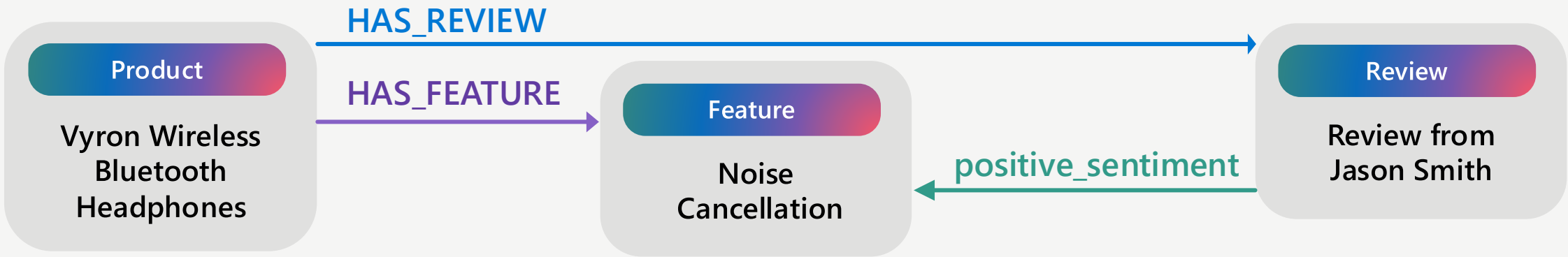
Turn relational and unstructured data into a queryable graph

Step 1: Build the graph with semantic operators



Use azure_ai.extract() to extract product features from review text and classify sentiment:

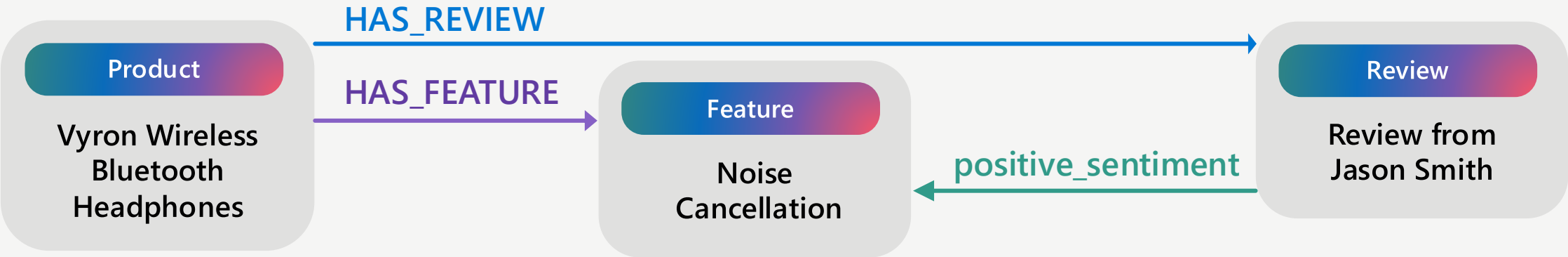
```
SELECT azure_ai.extract(f.review,
  ARRAY[f.feature_schema], 'gpt-5') ::JSONB->>'productFeature')
SELECT azure_ai.extract('Review:' || r.review || ' Feature:' || f.name,
  ARRAY['sentiment about feature-positive, negative, neutral'], 'gpt-5')
```



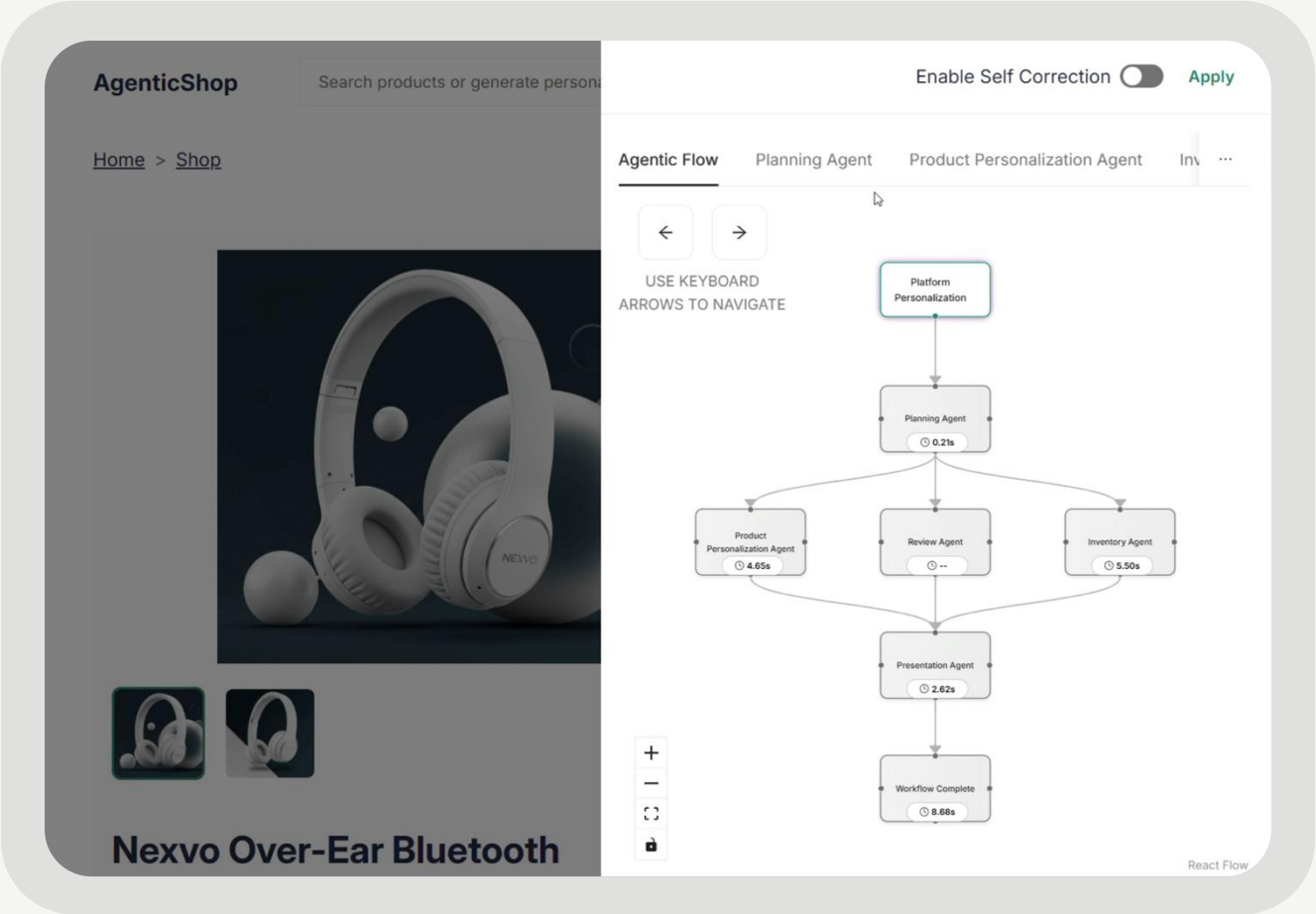
Step 2: Query the graph for complex relationships

Find reviews of headphones with positive sentiment about noise cancellation feature:

```
SELECT * FROM cypher('product_review_graph', $$
  MATCH (p:Product), (f:Feature), (r:Review)
  WHERE p.id IN {product_ids} AND f.name = 'noise cancellation'
  MATCH (p)-[frel:HAS_FEATURE]->(f)
  MATCH (f)<-[rel:positive_sentiment {{product_id: p.id}}]-(r)
  RETURN p.id AS product_id, COUNT(rel) AS positive_review_count $$) AS
graph_query(product_id agtype, positive_review_count agtype)
  ORDER BY (graph_query).positive_review_count DESC;
```



Demo: Agentic Shop with Azure Database for PostgreSQL



Code:
aka.ms/agentic-shop

Learn more Azure Database for PostgreSQL

Blog post: PostgreSQL + AI Agents
aka.ms/pg-ai-agents-blog

Learning path: Build AI apps with Azure Database for PostgreSQL
aka.ms/pg-ai-learn-path

Blog: Azure Database for PostgreSQL
aka.ms/azurepostgresblog

Install VS Code Extension for PostgreSQL
aka.ms/pgsql-vscode

Agentic retrieval with Azure AI Search

Users ask hard questions about unstructured data

Multiple questions in one query

What type of paint would be most suitable for a bathroom, and what is the estimated cost range for high-quality options?

Chained queries

Explain how to paint my house most efficiently. Then give me a list of the Zava products and prices for each supply

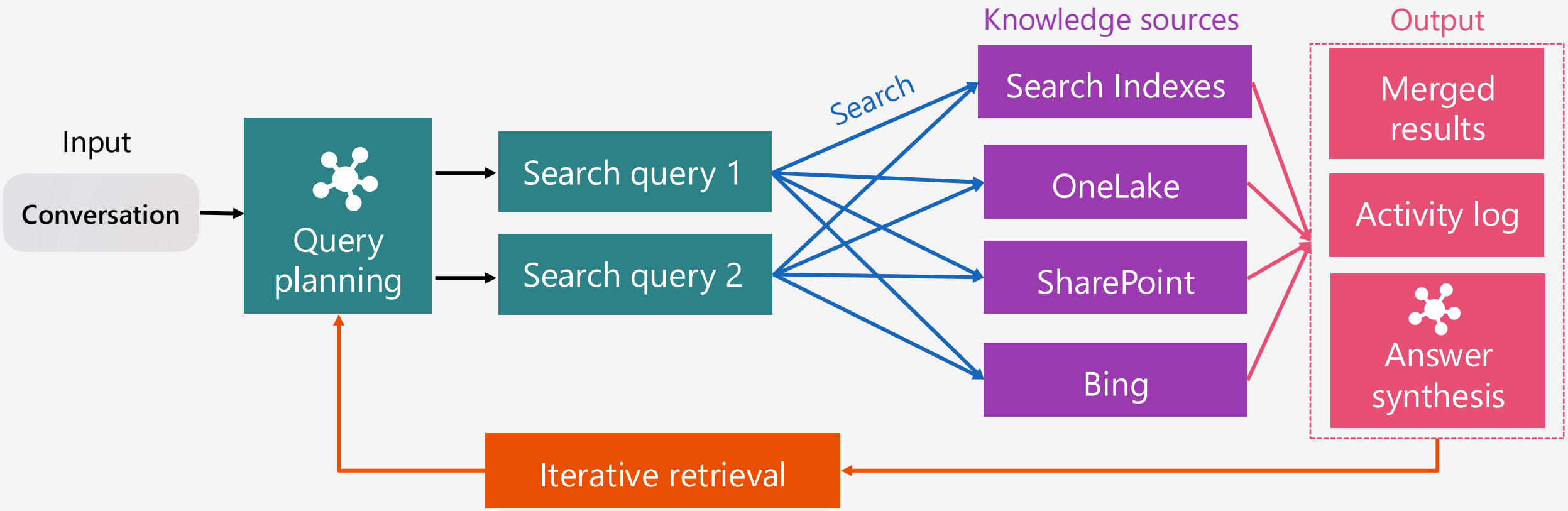
Queries requiring external knowledge

What's the best Zava sander for table corners and when do I replace my sanding pad?

Agentic retrieval with Azure AI Search knowledge base

Agentic methods applied to retrieval

Query planning	Knowledge source selection	Results merging



Search step uses hybrid search with semantic re-ranking when supported by the knowledge source

Knowledge base retrieval reasoning effort

MINIMAL

LOW

MEDIUM

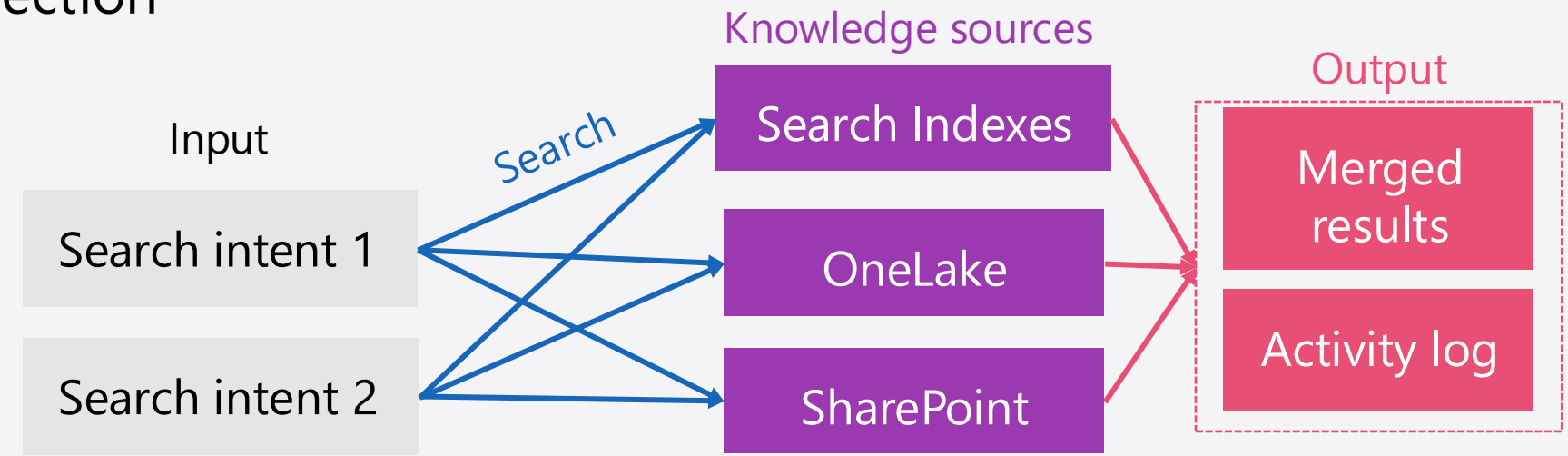
Low latency
Less model usage
Lowest cost

Highest quality
Increased stages
More agentic

Knowledge base with minimal effort

Minimal does *not* include:

- Query planning
- Knowledge source selection
- Web source
- Answer synthesis
- Iterative retrieval



Search intents represent queries.

All intents are sent to *all* sources.

Example of minimal effort

Input

"what's best Zava paint for bathroom walls?"

* You may optionally pass in *multiple* intents.

Output

Activity log

Step	Details
Search ""what's best Zava paint for bathroom walls?"	Source: Search index Results: 8
Search ""what's best Zava paint for bathroom walls?"	Source: SharePoint Results: 2

Intent is sent to *all* sources.

Merged results

.pdf

Interior Semi-Gloss Paint, price $47.0\r\n\r\n ## Brand\r\nZavaTech Hardware...

🏆 Re-ranker score: 2.95

ZavaBathroomPost.pdf

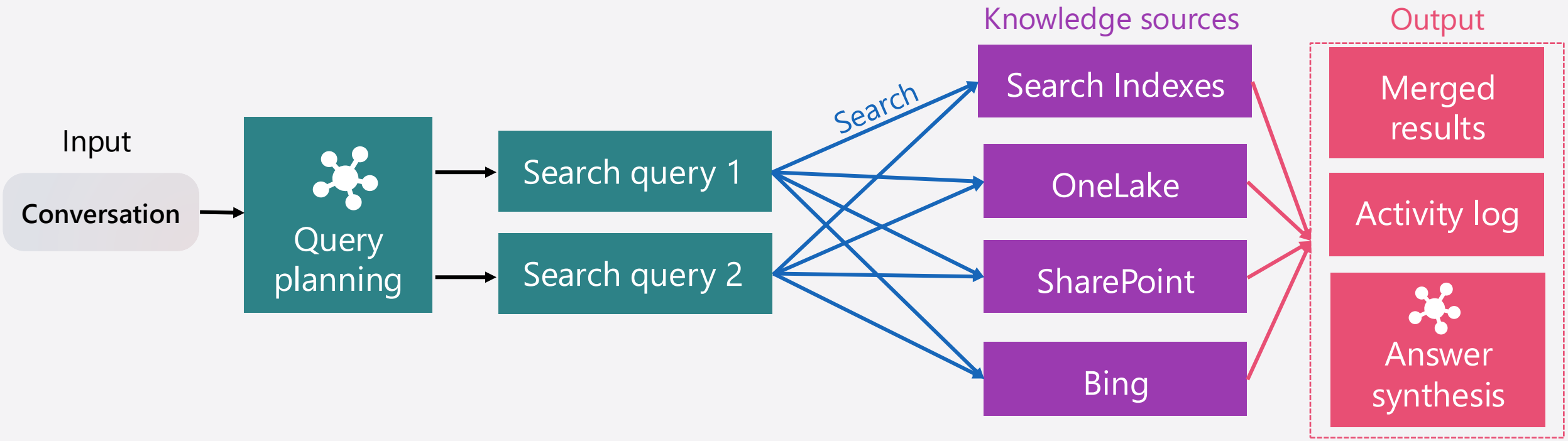
Refresh Your Bathroom with Confidence: Why Semi-Gloss Paint Makes All the Difference When it comes to painting projects, bathrooms are...

🏆 Re-ranker score: 3.07

Knowledge base with low effort

Low effort mode does *not* include:

- Iterative retrieval



Example of low effort with Bing web knowledge

Input

what's best Zava paint for bathroom walls, and how does it compare to other brand paints?

Output

Activity log

Step	Details
Query planning	Input tokens: 2113 Output tokens: 75
Search "paint for bathroom"	Source: manuals Results: 13
Search "paint for bathroom"	Source: Bing Web Results: 8
Search "cost of bathroom paint"	Source: manuals Results: 14
Search "cost of bathroom paint"	Source: Bing Web Results: 0
Agentic reasoning	Reasoning: 20K Effort: Low
Model answer synthesis	Input tokens: 7921 Output tokens: 108

Merged results

PFIP000003.pdf

Interior Semi-Gloss Paint, price $47.0\r\n\r\n ## Brand\r\nZavaTech Hardware...

🏆 Re-ranker score: 2.95

coolpaints.com

Url: https://www.coolpaints.com/bathroom-moisture-paint-issues-one-weird-trick

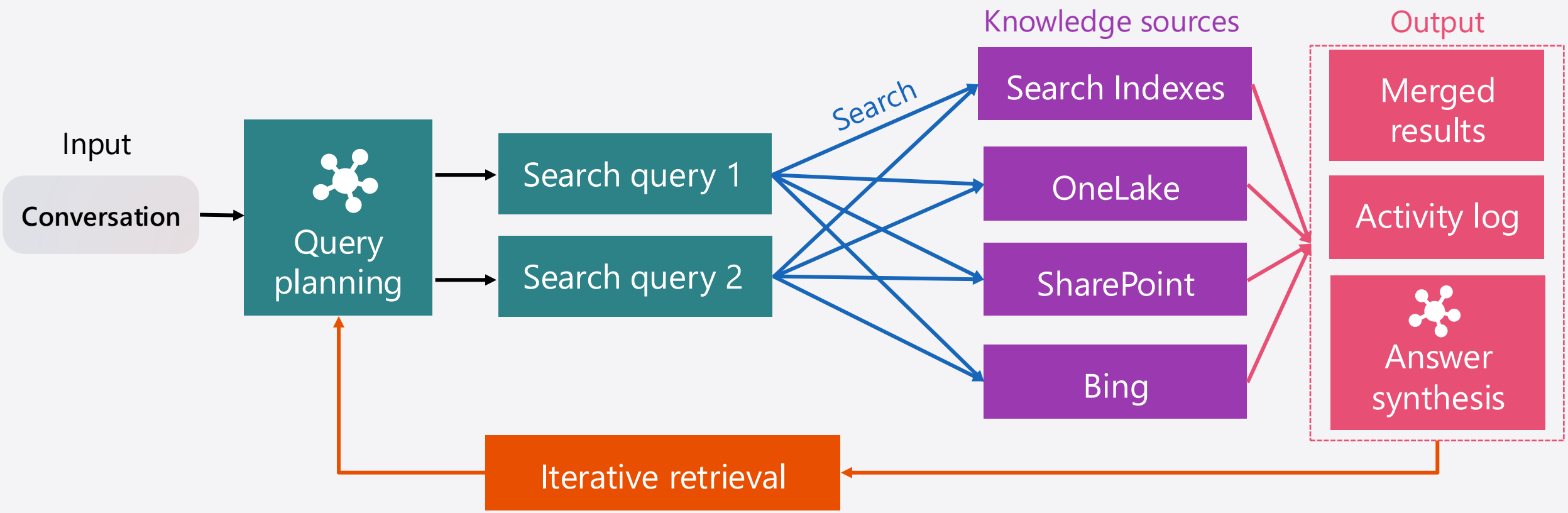
Title: Why semi-gloss paint is the best for bathrooms

Answer synthesis

A semi-gloss interior paint is suitable for use in a bathroom because it offers moisture resistance, is washable, and stands up well to humidity and frequent cleaning. This type of paint is specifically recommended for bathrooms, kitchens, and trim work due to its durability and ease of maintenance [ref_id:1]. The typical cost for a can of Interior Semi-Gloss Paint from ZavaTech Hardware Solutions is $47.00 [ref_id:2].

Answer synthesis is *required* for web knowledge.

Knowledge base with medium effort



Example of medium effort

Input

Explain how to paint my house most efficiently. Then give me a list of the Zava products and prices for each supply

Output

Activity log

Step	Details
Query planning	Input tokens: 1484 Output tokens: 115
Search "efficient house painting"	Source: manuals
Search "Zava paint supplies prices"	Source: manuals
Search "efficient house painting"	Source: web
Search "Zava paint prices"	Source: web
Query planning	Input tokens: 1169 Output tokens: 249
Search "paint brushes rollers trays"	Source: manuals
Search "paint brushes rollers trays"	Source: web
Agentic reasoning	Reasoning: 65K Effort: Medium
Model answer synthesis	Input tokens: 12676 Output tokens: 196

Merged results

PFEP000007.pdf

Exterior Acrylic Paint, price $57.0\r\n\r\n## Brand\r\nZavaTech...

🏆 Re-ranker score: 2.85

francoisestmoi.com

Url: http://francoisetmoi.com/diy/top-5-ways-to-paint-more-efficiently

Title: Top 5 Ways to Paint More Efficiently

qualitypreferred.com

Url: https://www.qualitypreferred.com/choosing-the-right-method-the-best-way-to-paint-a-house

Title: The Best Way to Paint a House

Answer synthesis

The most efficient way to paint a house is to work from the top down: start by prepping all surfaces, then paint ceilings first, followed by trim and baseboards, and finish with the walls, using rollers for large areas and brushes or pads for detail work to minimize drips and touch-ups [ref_id:5][ref_id:3][ref_id:4]. For exterior painting, clean and repair surfaces, prime exposed areas, and always paint from high to low, finishing with doors and trim [ref_id:5][ref_id:6].

Below is a list of Zava paint supplies:
* Exterior Acrylic Paint: $57.0 [ref_id:1]

Demo: Open source solution with agentic retrieval

Agentic retrieval response

deployment: null | filter: null | model: null | reranker_threshold: 1.9 | effort: medium

Iteration 1 Execution steps

Step	Details	Elapsed MS
Step 1: Query planning	Input: 2171 / Output: 140 / Total tokens: 2311	1645
Step 2: Index search	**Source:** zavadocs **Search:** most efficient way to paint a house *No results found*	801
Step 3: Index search	**Source:** zavadocs **Search:** Zava products for house painting supplies and their prices PFIP000003.md PFIP000002.md PFIP000001.md	213
Step 4: Web search	**Source:** web **Search:**most efficient way to paint a house How to Paint a House Quickly and Efficiently - The Spruce How To Paint Quickly and Efficiently - This Old House What Is the Correct Sequence of Painting a House? What is the fastest way to paint a house? - Resto NYC How to Paint a House: 12 Steps (with Pictures) - wikiHow How To Paint A House Interior Quickly - Interior Ideas 13 Top Tips to Make Any Paint Job Go Faster - Bob Vila How to Paint a Room (The Correct Order of Operations) Best Way To Paint a Room Fast: Quick Tips for a Smooth Finish The Most Efficient Way to Paint a Room | Brad the Painter	1058
Step 5: Web search	**Source:** web **Search:**Zava products for house painting supplies and their prices *No results found*	498

Code:
aka.ms/ragchat

Docs:
aka.ms/ragchat/knowledgebase

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aka.ms/AISearch-new

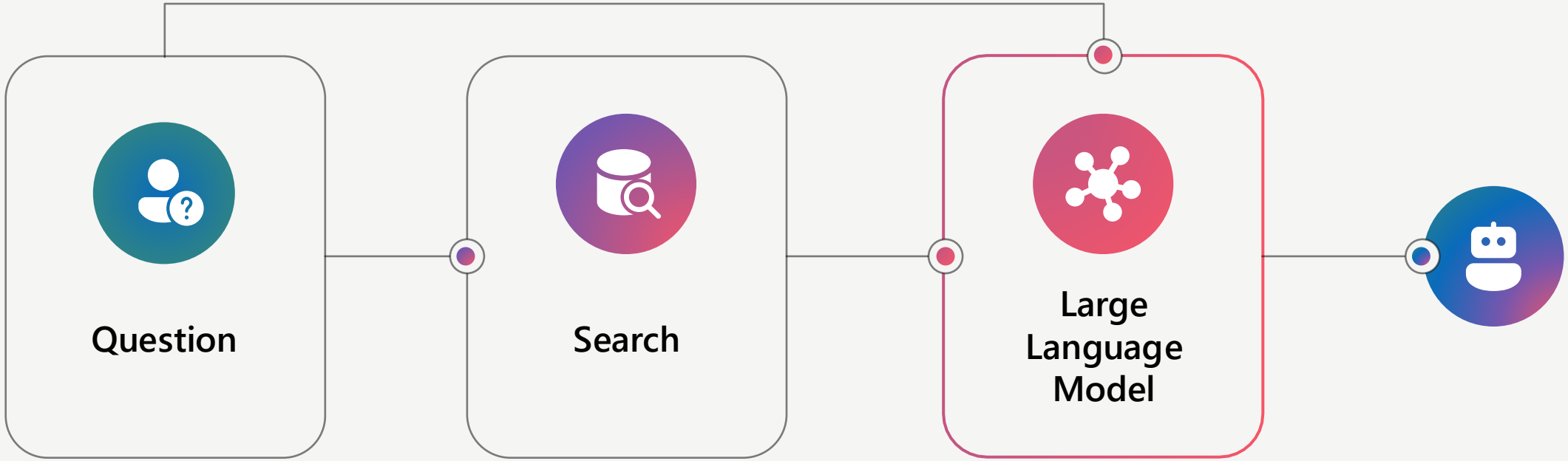
Foundry IQ Public Preview
aka.ms/FoundryIQ

GitHub Sample – Quickstart
aka.ms/AISearch-ar-pyn

What's next?

Design your agentic flows carefully

- Evaluate your agent at each step in the process
- Retrieval is often a big factor affecting output quality
- Data preparation is very important, as it affects retrieval
- Set up automated evaluations so that you can scientifically measure different search approaches, data preparation techniques, and models



Thank you!

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Q&A

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